[Date]

Business and Ethical Evaluation

BIN 381 Milestone 3

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Declaration of Academic Integrity

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understand that plagiarism is the act of using someone else's words, ideas, or work without proper attribution or acknowledgement, presenting them as one's own. It can involve copying and pasting text, paraphrasing without citation, or closely imitating the structure and ideas of another source without giving credit.

I understand that our Project must be accurately referenced and cited.
This Project is our own work. We acknowledge that copying someone else's work, or part of it, is wrong and that submitting identical work to others constitutes a form of plagiarism.

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of honesty and integrity as they relate to academic work.

AI was used to help debug R code and troubleshoot evaluation metrics. All findings, interpretations, and conclusions were made by the author based on the actual data and model results.

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TASK 3

3.1)

ProvinceCode <int>	.metric <chr>	.estimator <chr>	.estimate <dbl>
1	recall	binary	0.1636364
2	recall	binary	0.2608696
3	recall	binary	0.2727273
4	recall	binary	0.3333333
5	recall	binary	0.3509934
6	recall	binary	0.2784810
7	recall	binary	0.1301370
8	recall	binary	0.2450980
9	recall	binary	0.0500000

Description: df [9 × 2]

ProvinceCode <int>	n <int>
1	55
2	115
3	44
4	48
5	151
6	79
7	146
8	102
9	20

- The recall rate is quite varied among the provinces , ranging from 5.0% in Province 9 to 35.1 in Province 5 almost seven times difference. Recall is not a matter of sample size alone because there are 20 genuine food insecure households in Province 9 test set of which only 1 has been recalled by the model. Similar is the case with province 7 where there is a low recall rate(13.0%) and a larger sample size(146 households).

- This creates a problem of fairness since a tool that is used for the purpose of prioritizing intervention efforts would, based on current configuration of the model , serve the food insecure households in Provinces 9 and 7 much less than those in Province 5 . Without additional exploration(wether is because of a smaller number of training cases in these two provinces , or due to different predictor-food insecurity relationships , or some other reason) , this problem needs to be acknowledged

- By the technical success criteria set in Milestone 1 , the model performs well : ROC-AUC (0.840) comfortably exceeds the 0.75 threshold , more than the required 3 significant variables were identified , and both modelling approaches converge on the same key drivers (income , dwelling quality , province)

- However being able to meet the ROC-AUC threshold does not mean the model is ready to support the started business decision , Milestone 1's business objective is to help identify vulnerable households for target intervention and at the model's default classification threshold

intervention and at the model's default classification threshold , only 24.6% of genuinely food-insecure households are actually flagged. A high ROC-AUC reflects good ranking ability across all possible thresholds , but the specific threshold needed for practical deployment (catching enough at risk- households to inform real resource allocation) was not part of the original technical success criteria , and is arguably a gap is in how success was originally defined.

- A threshold increase to 0.3 brings the recall up to 63.3% better matching the business goal, but it does come with the trade-off of precision (48.8%), in that almost half of the identified cases would not actually experience food insecurity. This trade-off and optimal threshold for the business goal need to be an intentional decision.

3.2) Association vs Causation

* The variable identified as important such as income, dwelling quality , province , perceived income are associations that have been observed through the model analysis not as causative factors . For instance , individuals with low income face a high chance of being food insecure ; however the findings does nit prove that increasing the income level alone can reduce food insecurity , as other factors not considered in the analysis (for instance , unexpected illness, food costs , job instability) may affect both the independent variable and the dependent variables. R6 risk from Milestone 1 predicted this same risj and concluded that findings should be considered associations and not causation.

3.3)

A tibble: 4 × 4

HeadPopulationGroupCode <int>	.metric <chr>	.estimator <chr>	.estimate <dbl>
1	recall	binary	0.2553495
2	recall	binary	0.1296296
3	recall	binary	1.0000000
4	recall	binary	0.0000000

4 rows

A tibble: 2 × 4

HeadSexCode <int>	.metric <chr>	.estimator <chr>	.estimate <dbl>
1	recall	binary	0.2549505
2	recall	binary	0.2359551

2 rows

Description: df [4 × 2]

HeadPopulationGroupCode <int>	n <int>
1	701
2	54
3	1
4	4

4 rows

- Recall was also examined by HeadPopulationGroupCode to verify the differential model performances across population groups, as committed in Milestone 1 ethics log. Group 1(n=701) and 2 (n=54) have enough sample size to support a reliable comparison , and show a meaningful gap 25.5% recall for Group 1 versus 13.0% for Group 2 , the model catches roughly half as many food insecure households in Group 2 relative to Group 1 .

- Groups 3 (n=1) and 4 (n=4) shows extreme recall values (100% and 0% respectively), but these cannot be interpreted as an indication of performance discrepancy since they represent a small sample with numbers that are within the test sample . These results do not allow conclusions on the model's fairness towards these groups due to insufficient data.
- Together with the earlier province level disparity (recall rate ranging from 5.0% to 35.1%) , this shows that the accuracy of the model is detecting food insecure households varies . The disparity at the province level is more substantial since it has a larger sample size ; on the other hand , the disparity at the population group level between Groups 1 and 2 is a separate concern. Both should be paid attention before using the results for resource distribution as per Mitigation Milestone 1 ("will test model performance across demographic subgroups; use fairness metrics")

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3.4 Is the model useful and responsible enough to deploy?

The random forest ranks households well (ROC AUC 0.840, against the 0.75 target from Milestone 1). But ranking well isn't the same as being usable. What the business needs is to find most food-insecure households without too many false alarms, and to do it reasonably fairly across provinces. Table 1 checks the model against each of these needs.

Requirement	Evidence from the test set	Verdict
Tells food-insecure and secure households apart	ROC AUC 0.840 (logistic regression 0.819). Target was 0.75.	Met
Probabilities work as risk scores	Predicted and actual risk are close in most bands: 25% predicted vs 22% actual in the 0.2-0.3 band, and 77% vs 80% above 0.7. Risk is slightly overestimated below 0.1 (4.7% predicted, 2.5% actual).	Acceptable
Finds the households we want to reach	At the default 0.50 cut-off, recall is 24.6% (187 of 760). At 0.30 it is 63.3% (481 of 760).	Not met at 0.50, met at 0.30
False alarms are tolerable	At 0.30, precision is 48.8%. 985 households are flagged and 504 of them aren't food insecure. 279 food-insecure households are still missed. About 24% of households get flagged, against an 18% base rate.	OK for screening, not for individual decisions
Similar results across provinces	Recall at 0.30 runs from 73% (KwaZulu-Natal) down to 49% in Gauteng and 35% in Limpopo (only 20 cases; 95% interval roughly 18-57%).	Partly met, needs monitoring
Similar results across other groups	Recall at 0.30: head of household sex code 1 = 60%, code 2 = 67%; geography type 1 = 59%, type 2 = 70%. Population groups 3 and 4 have only 1 and 4 food-insecure households, so we can't judge them.	Small gaps; some groups can't be assessed
Results can be explained	Both models point to the same drivers: income, spending, perceived income, dwelling quality and province.	Acceptable, as associations
Data are current and representative	Trained on one survey year (2024), and the inputs only exist in survey-type data.	Limitation: revalidate each cycle

Table 1. Deployment readiness of the random forest.

Decision: deploy as decision support, with conditions

Recommendation: deploy the random forest as a screening and prioritisation tool for analysts, with a threshold of 0.30. Don't use it to make automatic decisions about individual households.

The threshold is a business choice, not a statistical one. Figure 1 shows the trade-off. At 0.30 the F1 score peaks (0.55), recall passes 60%, and about half the flags are correct. At 0.50 precision is better (68%) but three in four food-insecure households are missed, which defeats the Milestone 1 goal. At 0.20 we'd find 82% of them but flag 40% of all households, probably more than a programme could follow up.

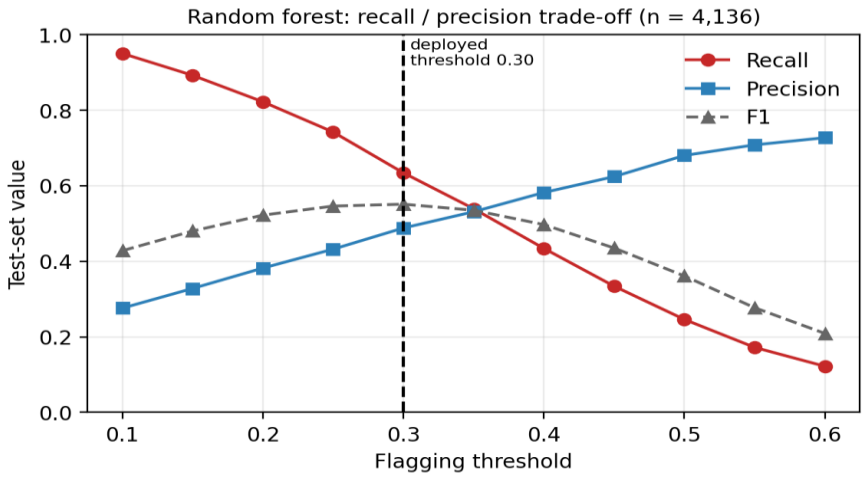


Figure 1. Recall, precision and F1 by threshold (random forest, test set).

Conditions for deployment:

1) Start at 0.30 and let the programme owner confirm it, since it reflects how costly a missed household is compared with a false alarm.
2) Use scores to prioritise areas and outreach lists, with a person reviewing every flagged case. Nobody is refused help because of a low score.
3) Show the Gauteng and Limpopo caveat in the app (done) and check those provinces first when results look odd. Province-specific thresholds might help, but we haven't tested them.
4) Monitor drift, data quality and province-level recall (section 4.5), and revalidate whenever a new survey year comes out.
5) Always describe drivers as associations, not causes (done in the what-if tab).

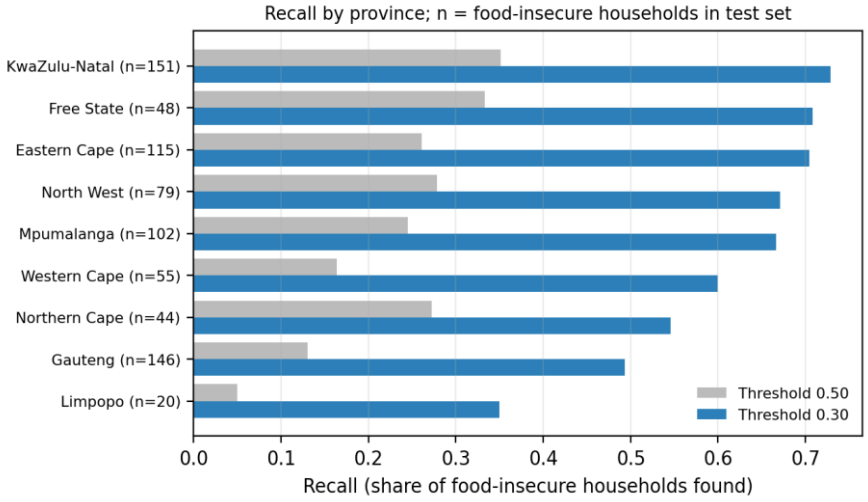


Figure 2. Recall by province at the default (0.50) and deployed (0.30) thresholds.

Dropping the threshold narrows the province gap Person B found at 0.50 (5% in Limpopo to 35% in KwaZulu-Natal) to 35%-73%, but doesn't close it. Limpopo also has a very low food-insecurity rate (about 4%, against 13-30% elsewhere), so the model had few positive cases to learn from there. Read its Limpopo scores with extra care.

A gap in our Milestone 1 criteria. We met the technical criteria (AUC of at least 0.75, at least three significant drivers), but none of them said how many at-risk households the tool must find. We suggest adding two operating criteria: recall of at least 60% at the deployed threshold, and recall of at least 40% in every province with 30 or more food-insecure households. The monitoring module checks both.

A small note: the app flags households at 0.30 or above, while Person B's table used strictly above 0.30. That changes one household (482 vs 481 correct flags), so it doesn't matter.

3.5 Ethical issues in deployment and how we handle them

These add to Person B's ethics log with issues that only show up once the model is in use. Each mitigation is built into the app or the monitoring plan.

Ethical issue	Why it matters here	Mitigation
Using scores for individual decisions	49% precision is not a safe basis for refusing someone support, and a low score doesn't prove a household is secure.	Warning banner in the app, a 'not intended for' list in the model card, and a required human review step.
Unequal accuracy across groups	Recall is lowest in Limpopo and Gauteng, so households there are more likely to be missed. Some population groups are too small to test.	Province recall check with WATCH/ALERT status, caveat shown to users, threshold review if a province drops below 40% recall, and more data for small groups.
Missed households and false alarms	A missed household goes without help. False alarms waste scarce resources and can stigmatise.	Threshold favours recall. The slider shows the trade-off. Flags are described as 'higher risk', not as a finding.
Privacy of household records (POPIA)	Uploads and downloads contain household-level survey data.	App stores nothing between sessions, download warning, host internally for real data, remove direct identifiers before upload, and limit access to named analysts.
Reading drivers as causes	The what-if tab could be read as 'raise income by R X and the household becomes secure'.	Disclaimer on the tab, and drivers always described as associations (Milestone 1 risk R6).
Over-relying on the score	People may treat a probability as a fact.	Risk bands, a reminder of the base rate (about 18%), and local knowledge used when reviewing flags.
Quiet deterioration	Prices, grants and incomes change, so an old model can keep giving confident but wrong scores.	Drift and performance monitoring with retrain and retire rules (section 4.5).
Being open about AI use	The assignment requires AI use to be acknowledged.	Entry in the responsible AI usage log at the end of this section.

Table 2. Deployment-stage ethical issues and mitigations.

Task 4 - Deployment

4.1 Deployment strategy

Element	Design
Intended users	Analysts and programme planners in provincial or municipal departments and NGOs working on food security. The BI analyst maintains the model, and programme managers read the summaries.
Decisions supported	Which areas and household groups to prioritise for food-relief outreach and eligibility checks, and how long the follow-up list should be. The tool doesn't decide eligibility.
Inputs	A CSV with one row per household and the same fields as the Milestone 2 dataset, plus a flagging threshold (default 0.30). Outcome labels (FoodInsecure_flag) are optional and switch on the performance checks.
Outputs	Per household: probability of food insecurity, flag and risk band (Lower, Elevated, High). Per batch: counts, share flagged, risk distribution, flags by province, a downloadable scored CSV, and monitoring status.

Table 3. Deployment design.

How it works on a daily (Figure 3):

1) An analyst gets household data in the agreed format (a new survey release or similar extract).
2) They upload the CSV to the Shiny app and pick a threshold.

3) The app validates the file and reports any fixes or problems before scoring.
4) The random forest scores each household, and the app flags and bands them.
5) The analyst reviews the flagged households (keeping the province caveats in mind) and downloads the list.
6) The programme owner decides what to do. Scores support the decision but don't replace it.
7) The Monitoring tab (or run_monitoring.R) runs on every batch, and again when outcomes are known. Any WATCH or ALERT goes to the model owner for a retrain or retire review.

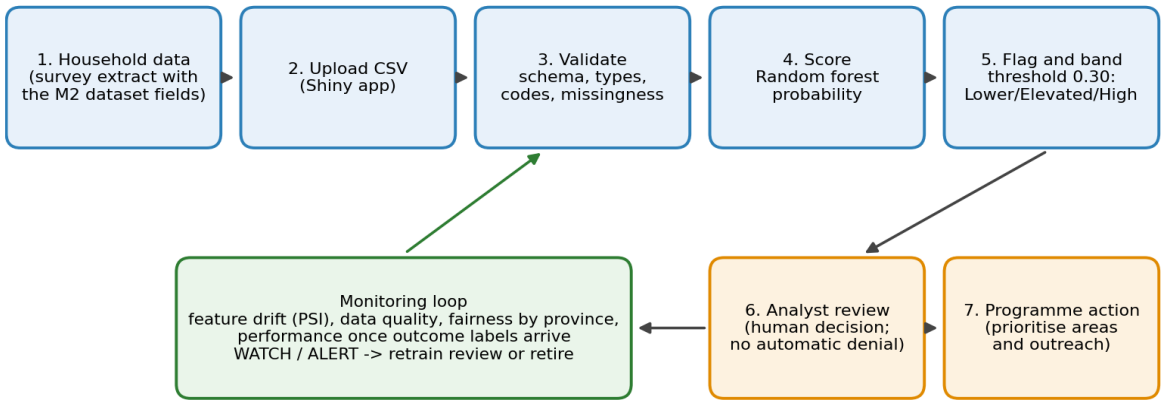


Figure 3. Workflow and monitoring loop.

4.2 Comparing deployment tools

The brief asks for an R-based tool, so we left out Python and BI options like Streamlit and Power BI and compared four R options.

Criterion	R/Shiny	Quarto / R Markdown report	Quarto or flexdashboard dashboard	plumber REST API
Uses the saved R model without re-coding	Yes	Yes	Yes	Yes
Interactive upload, threshold and what-if	Yes, full control	No (parameters at render time only)	Limited unless a Shiny runtime is added	No interface; needs a client
Usable by non-programmers	High	Medium (read-only)	Medium to high	Low
Shows validation messages and monitoring	Yes, live	Static snapshot	Partly	Returns data only
Hosting effort	Medium	Low	Low (static) to medium	Medium to high
Fit for this project	Best	Good for periodic reports	Possible	Good as a later back-end

Table 4. Comparison of R deployment options.

Recommendation: R/Shiny**.** It's the only option that gives us upload, validation feedback, a threshold control and monitoring in one tool, and it reuses the saved tidymodels workflow, so scoring in the app matches the modelling notebook exactly. A Quarto report would work well for periodic monitoring summaries, and a plumber API could be added later if another system needs to call the model.

Hosting. Because the inputs are household-level records, real use should be hosted inside the organisation (Posit Connect, Shiny Server or an internal Docker container). A public cloud service is fine for a demo with the bundled data, but not for real household data without a POPIA assessment.

4.3 The deployed app

The Shiny app (app.R) has four tabs. It loads the final random forest (data/final_rf.rds), the Milestone 2 training data as a drift reference, and the test predictions as baseline scores.

1) Score households: upload data or use the demo set, pick the threshold, then validate and score. Shows validation notes, counts, a risk histogram, flags by province, the highest-risk households and a CSV download.

2) What-if household: a single household built from typical values with eight editable inputs (province, income, household size, head's age, comparative income rating, expenditure band, roof condition, pay-TV asset). For exploring and explaining only.

3) Monitoring: performance against baseline, recall by province, feature drift (PSI), data-quality checks and an automatic retrain-review message.

4) Model card: intended use, what it's not for, test performance, the threshold trade-off and known limitations.

A few design choices worth noting. Cleaning and scoring code is shared with the modelling notebook through preprocess.R, so nothing is duplicated. Learned preprocessing (imputation, dummy coding, scaling, correlation filter) lives inside the saved workflow, so new data is treated exactly as the training data was and nothing is re-learned from it. And probabilities are computed once per upload, so moving the threshold slider is instant.

4.4 How new data is validated and processed

Step	What happens	Outcome
1. Schema check	The file is compared with the predictors the model was trained on (read from the saved recipe).	Missing essential columns stop the run with a message naming them. Missing columns the recipe drops anyway are filled with NA.
2. Extra columns	Columns the model doesn't use are removed.	Reported as 'n extra columns ignored'.
3. Type check	Numeric predictors that arrive as text are converted.	Reports how many new missing values the conversion created.
4. Fixed cleaning rules	Codes 9 and 99 (unknown) become missing, and nominal survey codes become factors.	Same rules as in training (preprocess.R).
5. Quality warnings	Columns with over 50% missing values are listed.	Warning that predictions are less reliable.
6. Learned preprocessing	Inside the workflow: imputation with training medians and modes, new categories mapped to 'new', rare levels pooled, dummy coding, correlation and variance filters, standardisation.	Nothing is fitted on new data, so there's no leakage.
7. Scoring	Random forest probability of food insecurity, then flag and band from the threshold.	Table, charts and CSV download.
8. Post-scoring checks	The monitoring module compares the batch with the training data (PSI), checks unseen categories and missingness, and checks performance if labels are present.	OK / WATCH / ALERT statuses.

Table 5. Validation and processing of new data.

One gap: the validation doesn't check value ranges yet (for example a head of household aged 200). Odd values would go straight through to the recipe, so range checks are on the next-steps list.

4.5 Monitoring strategy

Baselines are the random forest's test-set results at 0.30. The WATCH and ALERT limits are our own proposed operating rules, set a sensible distance from the baseline, and should be reviewed whenever

the model is retrained. For scale, the 95% interval around the 63% recall baseline is roughly 60-67%, so the limits sit well outside normal noise. Province checks only apply where a batch has at least 30 food-insecure households in that province.

Measure	Baseline	WATCH	ALERT	Action on ALERT
ROC AUC (needs labels)	0.840	below 0.80	below 0.75	Retrain review (0.75 is the Milestone 1 criterion)
Recall at 0.30 (needs labels)	63%	below 55%	below 50%	Review threshold, then retrain
Precision at 0.30 (needs labels)	49%	below 40%	below 35%	Review threshold and data quality
Share of households flagged	24%	outside 19-29%	outside 15-33%	Look for drift in inputs or scoring
Province recall (needs labels)	35-73%	below 50%	below 40%	Fairness review; test a province-specific threshold
Feature drift, PSI of 11 key drivers	near 0	0.10-0.25	0.25 or more	Three or more drivers in ALERT triggers a retrain review
Score distribution PSI	0	0.10-0.25	0.25 or more	Retrain review
Unseen categories (per coded field)	0%	any	above 2%	Check codes; update the model if the new categories are real
Missing share in key drivers	low	above 10%	above 20%	Fix the data feed before scoring

Table 6. Monitoring measures and limits. PSI of 0.25 or more is the usual 'major shift' level (Siddiqi, 2006).

Schedule

. Every new batch: validation, data-quality, drift and share-flagged checks (Monitoring tab or run_monitoring.R, which logs each run to monitoring_logs/log.csv).

. When a new survey year is out and outcomes are known: full performance and province-recall check, then a retraining decision.

. Once a year: review the threshold with the programme owner, the monitoring limits and the ethics log.

Outcome labels only arrive when a new survey is published, so performance checks can only run about once a year. In between, the drift and quality checks act as early warnings.

When to retrain

. AUC below 0.75, or recall at the deployed threshold below 50%, on a labelled batch.

. Three or more key drivers (or the score distribution) at PSI 0.25 or more.

. A new survey release that changes the questions, codes or categories behind the model inputs.

. A province falls below the fairness limit and a threshold change can't fix it.

. Retraining means rerunning the pipeline from Milestone 2 with a fresh split, and passing the same comparison and ethics checks as this milestone before the new model replaces the old one.

When to retire

. Retraining can't get AUC back above 0.75.

. The official definition or measurement of food insecurity changes, so the target is no longer comparable.

. Key input variables are discontinued and can't be replaced.

. An ethics or fairness review finds harm that mitigation can't remove.

. The programme stops using the scores, or a better-validated tool replaces the model.

Roles

The model owner (BI analyst) runs the checks, keeps the log and proposes retraining. The programme owner confirms the threshold and decides what to do after an ALERT. The data steward protects uploaded data and approves access. Each retrained model is saved with a date in its file name, the previous one is kept so we can roll back, and every change goes in the log.

4.6 Repository and reproducibility

File	Purpose
app.R	Shiny interface and server logic
preprocess.R	Fixed cleaning rules, validate_new_data(), score_households() (shared with Task1.Rmd)
monitoring.R	PSI, data-quality, performance, fairness and retrain-rule functions
run_monitoring.R	Command-line monitoring with a run log
data/	final_rf.rds, training and test data, test predictions, threshold_table.csv
README.md, install_packages.R	Setup, run instructions, input requirements

Table 7. Deployment repository.

What we checked: We ran the monitoring functions on the saved test predictions and they reproduce the baseline (AUC 0.840, recall 0.634, precision 0.488, 23.9% flagged). We also simulated a 40% drop in household income: the PSI for income per capita rose to 0.265 (ALERT) and for household income to 0.237 (WATCH), so the drift check does pick up a material shift.

Limitations and next steps: Add range checks to the validation. The app has no sign-in and relies on the hosting platform for access control. Province-specific thresholds and a regularised or class-weighted model could be tested to narrow the province gap. And an renv lock file should be committed so package versions match the saved model.

Responsible AI usage log – Person C entry

Item	Entry
Tool	Claude (Anthropic), used as an assistant in October 2026.
Used for	Drafting the R code for the Shiny app, monitoring functions and batch script; analysing the saved test predictions (threshold, calibration and subgroup tables, confidence intervals).
Checked and decided by me	Check that the app was run with the saved model and works; that the figures match the group's results; that the threshold and monitoring limits were agreed with the group; that the text was edited well and referenced.
Risks considered	AI-written code may contain errors; AI-suggested limits are proposals and not facts; the text must be in my own words and checked against the group's results before submission.

References

Protection of Personal Information Act 4 of 2013 (South Africa).

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